



### RADIO WAVE

A type of electromagnetic radiation that has the longest waves. It travels far and very quickly and can be used to carry information, such as music.

### REACTION

A force in response to an opposing force.  
*See also* chemical reaction

### REFLECTION

The way light, sound, or other types of energy bounce back from a surface.

### RENEWABLE ENERGY

A type of energy that comes from a source will not run out, unlike energy from fossil fuels. Types of renewable energy include wind power, wave power, and solar power.

### SALT

A chemical formed when an acid reacts another substance, such as a metal or an alkali. The word is also used to specifically describe sodium chloride.

### SATELLITE

An object in space that travels around another in an orbit. Many satellites are human-made.

### SENSOR

A device or component of a machine that picks up information from its surrounding environment, such as changes in heat or light levels, and responds accordingly.

### SOFTWARE

The programs or set of instructions or data that run on a computer and control how it works, including the operating system and applications.



### SOLAR SYSTEM

The Sun and all the bodies held in orbit around it by the Sun's gravity. These bodies include planets, dwarf planets, moons, asteroids, and comets.

### SOLID

A state of matter where an element's atoms are joined together in a rigid structure. Solids are firm to the touch and have a definite shape rather than taking on the shape of their containers.

### SOLUTE

A substance that dissolves in a solvent to form a solution.

### SOLUTION

A mixture in which the molecules of a solute are evenly spread out among the molecules of a solvent. Solutions can be separated in several ways.

### SOLVENT

A substance (usually a liquid) in which a solute dissolves to form a solution.

### STEAM

Very small but visible water droplets suspended in the air, caused by the condensation of water vapor.

### STIMULUS

Something that encourages activity or response in people or things.

### STREAMLINED

Smoothly shaped, usually curved, to move easily through air or water.

### SUBLIMATION

The process by which a solid turns directly into a gas without first turning into a liquid.



### SUGAR

A usually sweet-tasting substance needed by cells to live and grow.

### SURFACE TENSION

An effect caused by the strong attraction of water molecules to each other that makes the surface of water behave like a stretched elastic membrane.

### SYNTHETIC

A word describing something that is human-made or that does not exist naturally.

### TISSUE

A group of similar cells that carry out the same function, such as muscle tissue, which can contract.

### TURBINE

A device with rotating fan blades that are driven by the pressure of gases, liquids, or steam. It converts energy into a different form. Turbines powered by the wind or by moving water are often used to generate electricity.

### UPTHRUST

The upward force exerted by a liquid or a gas on an object immersed in it.

### VACCINATION

Giving a vaccine to cause future immunity or protection against infectious diseases.

### VACCINE

A substance given to people to produce immunity against a specific infectious disease.

### VOLCANO

An opening in Earth's crust through which magma (liquid rock) erupts, and the resulting structure created by the eruption.



### VOLUME

The amount of space an object takes up.

### WATER PRESSURE

The amount of pressure exerted by water. The pressure increases with depth because of the weight of water above it. So the deeper you dive in the sea, the higher the pressure gets.

### WATER VAPOR

The gas that is produced when molecules of liquid water evaporate.

### WEIGHT

The force exerted on matter by gravity. The more mass an object has, the greater its weight.

### YEAST

A tiny single-celled fungus that feeds on sugars, producing carbon dioxide gas and alcohol.

#### Abbreviations used in this book

|     |                                                |
|-----|------------------------------------------------|
| /   | per—for example, km/h means kilometer per hour |
| cm  | centimeter                                     |
| °C  | degrees Celsius                                |
| °F  | degrees Fahrenheit                             |
| ft  | foot                                           |
| g   | gram                                           |
| in  | inch                                           |
| kg  | kilogram                                       |
| km  | kilometer                                      |
| lb  | pound                                          |
| m   | meter                                          |
| mph | miles per hour                                 |
| MYA | million years ago                              |
| oz  | ounce                                          |